**SIMATS ENGINEERING**
**ASSESSMENT TOOL 3**

**COURSE NAME: Artificial Intelligence** **COURSE CODE: CSA1706**

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**COURSE OUTCOME COVERED**

***CO3:*** *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

*Assessment Tool Weightage: 20%*

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**Time Duration: 1 Hour**
**QUESTIONS: 5** **Total Marks:30**

Annexure A

**Resolution Algorithm Implementation**

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***Code of Conduct:***

*I SR Kaarthika(192412294) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

Annexure A

**Resolution Algorithm Implementation**

## Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

**Goal**

Prove that **the ground is wet** using the Resolution Algorithm.

**Tasks**

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

**Negate the goal.**

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause (□) if the goal is proved.

## Question 2 – Student Assignment Submission

**Knowledge Base**

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

**Goal**

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

**Negate the goal.**

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

## Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

**Goal**

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

**Tasks**

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

## Question 4 – Placement Eligibility

**Knowledge Base**
1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

**Goal**
1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks
1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

## Question 5 – Access Control System

**Knowledge Base**
1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

**Goal**

Prove that **the user is granted access** using the Resolution Algorithm.

**Tasks**
1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (□).
5. Explain each resolution step and write the final conclusion.

1.

**RUBRICS FOR CALCULATION**

| Numerical Problems | | | | | |
|---|---|---|---|---|---|
| **Criteria** | **Weightage** | **Level 4 (Exemplary)** | **Level 3 (Proficient)** | **Level 2 (Developing)** | **Level 1 (Beginning)** |
| Problem Understanding | 20% | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| Method Selection | 20% | Selects most appropriate method/formula with strong justification | Selects correct method with minor errors | Inappropriate or partially correct method | Unable to select method |
| Solution Process | 25% | Logical, systematic steps with correct approach throughout | Mostly correct steps with minor mistakes | Disorganized steps; multiple errors | No clear process |
| Accuracy of Calculation | 20% | All calculations accurate; correct final answer | Minor calculation errors | Frequent errors; incorrect final answer | No valid calculations |
| Interpretation of Results | 15% | Clearly interprets results with units and engineering relevance | Basic interpretation with minor omissions | Limited or unclear interpretation | No interpretation |

**Q1. Rain and Wet Ground**
**1. Propositional Logic**
Let:

- **R** = It is raining.
- **W** = Ground is wet.

Given:

1. If it rains, the ground becomes wet:
   **R → W**
2. It is raining:
   **R**

Goal:
**W**
**2. Convert into CNF**
The implication:
**R → W**
is equivalent to:
**¬R ∨ W**
Therefore, the clauses are:

- **C1: ¬R ∨ W**
- **C2: R**

**3. Negate the Goal**
Goal = **W**
Negated goal:
**¬W**
So we add:

- **C3: ¬W**

**4. Resolution Steps**
Resolve C1 and C2:
**C1:** ¬R ∨ W
**C2:** R
Resolving **¬R** and **R**:
**C4: W**
Now resolve C4 with C3:
**C4:** W
**C3:** ¬W
Therefore:
**□ (Empty Clause)**
**5. Conclusion**
Since the empty clause □ is derived, the negation of the goal leads to a contradiction.
**Therefore, the goal W is proved.**
**Ground is wet.**

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**Q2. Student Assignment Submission**
**1. Propositional Logic**
Let:

- **S** = Rahul submitted the assignment.
- **M** = Rahul receives internal marks.

Given:

1. If a student submits the assignment, the student receives internal marks:
   **S → M**
2. Rahul submitted the assignment:
   **S**

Goal:
**M**
**2. CNF**
Convert the implication:
**S → M**

Therefore:
**¬S ∨ M**
Clauses:

- **C1: ¬S ∨ M**
- **C2: S**

**3. Negate the Goal**
Goal = **M**
Negated goal:
**¬M**
So:

- **C3: ¬M**

**4. Resolution**
Resolve C1 and C2:
**C1:** ¬S ∨ M
**C2:** S
Therefore:
**C4: M**
Now resolve C4 and C3:
**C4:** M
**C3:** ¬M
Therefore:
□
**5. Conclusion**
The empty clause is obtained.
**Therefore, Rahul receives internal marks.**
The goal is **proved**.

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**Q3. Library Membership**
**1. Propositional Logic**
Let:

- **L** = Priya is a library member.
- **B** = Priya can borrow books.

Given:

1. If a person is a library member, then the person can borrow books:
   **L → B**
2. Priya is a library member:
   **L**

Goal:
**B**
**2. CNF**
Convert:
**L → B**
to:
**¬L ∨ B**
Clauses:

- **C1: ¬L ∨ B**
- **C2: L**

**3. Negate the Goal**
Goal = **B**
Negated goal:
**¬B**
So:

- **C3: ¬B**

**4. Resolution**
Resolve C1 and C2:
**¬L ∨ B**
**L**

Therefore:
**C4: B**
Now resolve:
**C4: B**
**C3: ¬B**
Therefore:
□
No further clauses are required because the empty clause has been obtained.
**5. Conclusion**
Since □ is obtained, the negation of the goal is contradictory.
**Therefore, Priya can borrow books.**
The goal is **proved**.

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**Q4. Placement Eligibility**
**1. Propositional Logic**
Let:
- **A** = Arun clears the aptitude test.
- **P** = Arun is eligible for placement.

Given:
1. If a student clears the aptitude test, then the student is eligible for placement:
   **A → P**
2. Arun cleared the aptitude test:
   **A**

Goal:
**P**
**2. CNF Clauses**
Convert:
**A → P**
to:
**¬A ∨ P**
Therefore:
- **C1: ¬A ∨ P**
- **C2: A**

**3. Negate the Goal**
Goal:
**P**
Negated goal:
**¬P**
Therefore:
- **C3: ¬P**

**4. Resolution Steps**
Resolve C1 and C2:
**C1:** ¬A ∨ P
**C2:** A
Resolving **¬A** and **A**:
**C4: P**
Now resolve C4 and C3:
**C4:** P
**C3:** ¬P
Therefore:
□
**5. Resolution Tree**
```
¬A ∨ P      A
   \       /
    \     /
       P
       |
```

¬P
|
□

**6. Conclusion**
The empty clause □ is derived.
Therefore, the negated goal **¬P** is false.
**Hence, Arun is eligible for placement.**

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**Q5. Access Control System**
This question has **two implications**, so resolution is performed in two stages.

**1. Propositional Logic**
Let:

- **P** = User enters the correct password.
- **A** = User is authenticated.
- **G** = User is granted access.

Given:

1. If the user enters the correct password, then the user is authenticated:
   **P → A**
2. If the user is authenticated, then the user is granted access:
   **A → G**
3. User entered the correct password:
   **P**

Goal:
**G**

**2. Convert into CNF**
For the first implication:
**P → A**
becomes:
**¬P ∨ A**
For the second implication:
**A → G**
becomes:
**¬A ∨ G**
Fact:
**P**
Therefore:

- **C1: ¬P ∨ A**
- **C2: ¬A ∨ G**
- **C3: P**

**3. Negate the Goal**
Goal:
**G**
Negated goal:
**¬G**
Add:

- **C4: ¬G**

**4. Resolution Step 1**
Resolve C1 and C3:
**C1:** ¬P ∨ A
**C3:** P
Resolving **¬P** and **P**:
**C5: A**

**5. Resolution Step 2**
Resolve C2 and C5:
**C2:** ¬A ∨ G
**C5:** A
Resolving **¬A** and **A**:

**C6: G**

**6. Resolution Step 3**

Resolve C6 with C4:

**C6:** G

**C4:** ¬G

Therefore:

□

**7. Resolution Tree**

```
 ¬P ∨ A      P
     \      /
      \    /
        A
        |
     ¬A ∨ G
        |
        G
        |
       ¬G
        |
        □
```

**8. Final Conclusion**

The empty clause □ is derived. Hence, the negation of the goal produces a contradiction.

**Therefore, the user is granted access.**

The goal **G is proved using the Resolution Algorithm.**

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